

9/7/26

TASK-3

- * Create a structure variable of a student estimate the size and structure manually. then
- Write a C program to find the size of structure now using size of operators. now, observe both the outputs. Are they same or different? Justify your answer.

C programming.

```
#include <stdio.h>
```

```
struct student
```

```
{
```

```
    int id;
```

```
    char grade;
```

```
    float marks;
```

```
};
```

```
int main()
```

```
    struct student s;
```

```
    printf("Size of
```

```
structure student = %d\n",
```

```
sizeof(s));
```

```
}
```

manual size calculation.

Data member	Data type	size (bytes)
id	int	4
grade	char	1
marks	float	4
Total		9 bytes

Actual size using size of

Size of structure student = 12 bytes.

Why are they different?

The outputs are different.

- Manual Size : 9 bytes (4+1+4)
- Size of result : 12 bytes (on most compilers)

This difference occurs because the compiler adds padding bytes b/w members to align data in memory for faster access.

Memory layout.

member	Size	Offset
id	4 bytes	0-3
grade	1 byte	4
padding	3 bytes	5-7
marks	4 bytes	8-11

So, the total size become 12 bytes instead of 9 bytes.

Hence, they are different because the compiler inserts padding to satisfy memory alignment requirements, improving performance.
The exact size may vary depending on the compiler and system architecture.